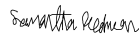


Solar Stirling Cycle Final Project

We hereby certify that the assignment herein was completed in accordance with the McKelvey School of Engineering and Undergraduate Student academic integrity policies. We acknowledge that artificial intelligence (AI) tools were used to generate some content in this report.

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Executive Summary.

The objective of this project is to analyze a solar-powered energy system to meet the electrical demands of the WashU machine shop. The system uses a solar dish to concentrate sunlight onto a single point called a receiver. This sunlight creates heat to drive a Stirling engine, a specialized motor that converts thermal energy into mechanical movement, which is then transformed into electricity by a generator. Additionally, the design incorporates a battery storage system to ensure a steady power supply, even after sunset or during cloudy periods.

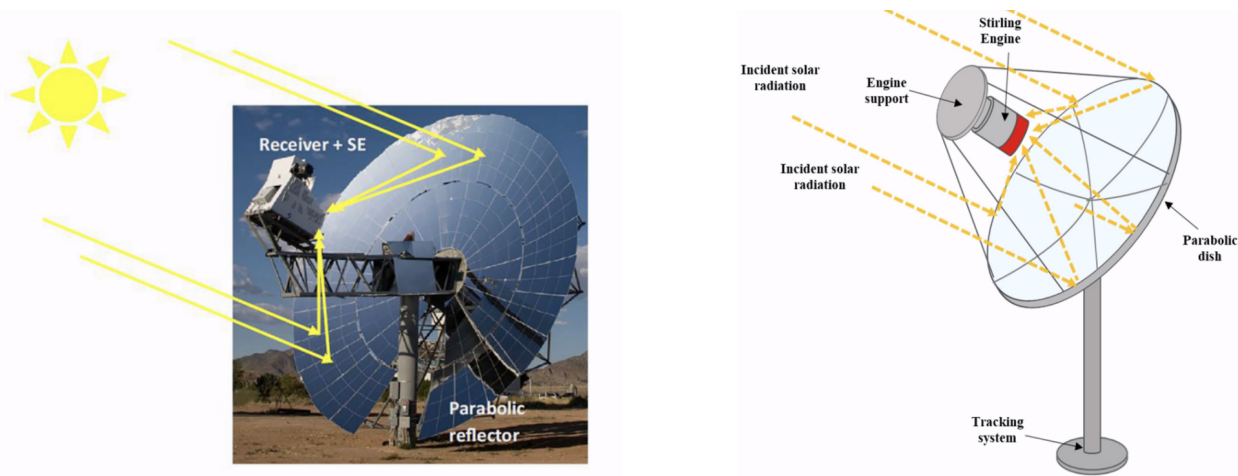
The results indicate that the system can generate enough power to meet the shop's expected load. Cooling fins could also be integrated into the system to maintain sufficiently low engine surface temperatures. The system produces a surplus of energy during daylight hours of both winter and summer months, which is stored in a battery bank to maintain operations at night.

Introduction.

The growing energy demand and increasing environmental problems has led to further investigation of clean energy. Solar energy is one of the most viable clean energy sources for power production, as it reduces fossil fuel consumption and decreases pollutants. The use of concentrated solar power through dish-Stirling systems was applied by integrating solar concentrators to the Stirling engines. The dish-Stirling systems first convert thermal energy into mechanical energy using concentrators and a Stirling engine, then the mechanical energy is converted to electricity using generators.

The solar dish-Stirling system has been proven as the most efficient energy generation method using solar energy. The system converts about 30% of the sun's radiation into electrical energy. The systems are modular and are self-contained power generators. Therefore, they can be installed to power plants and specifically in this project to power the machine shop in the Urbauer building on the WashU campus. For performance analysis of a dish-Stirling system parameters like concentration ratio, absorber temperature, host side temperature, cold side temperature, regenerator effectiveness, working fluid, dead volume, and average working pressure values are considered [1].

A solar dish Stirling system is comprised of the parabolic collector arrangement, a Stirling engine, and a power generator located at the focus of the dish. The Dish Stirling System is pictured in Fig. 1.



(a) Dish Stirling System and Components [1].

(b) Schematic View of Dish Stirling System.

Figure 1 Dish Stirling system schematics.

The collector system has 2 central parts: a solar concentrator and a thermal receiver. the collector system delivers thermal energy that runs the Stirling engine. The parabolic dish focuses the solar radiation on the receiver's aperture. The receiver is made of an aperture and an absorber. The aperture takes in the solar radiation delivered by the dish, and the absorber then transfers the thermal energy to the working gas within the Stirling engine. Then the Stirling engine converts the thermal energy into mechanical, which then becomes electrical [1].

The engine operates on the Stirling cycle which is a closed regenerative thermodynamic cycle. It consists of two isochoric and two isothermal processes. The ideal Stirling cycle includes a constant temperature expansion, constant volume heat addition, constant temperature compression, and constant volume heat rejection [2]. An ideal T-S diagram is shown in Fig. 2.

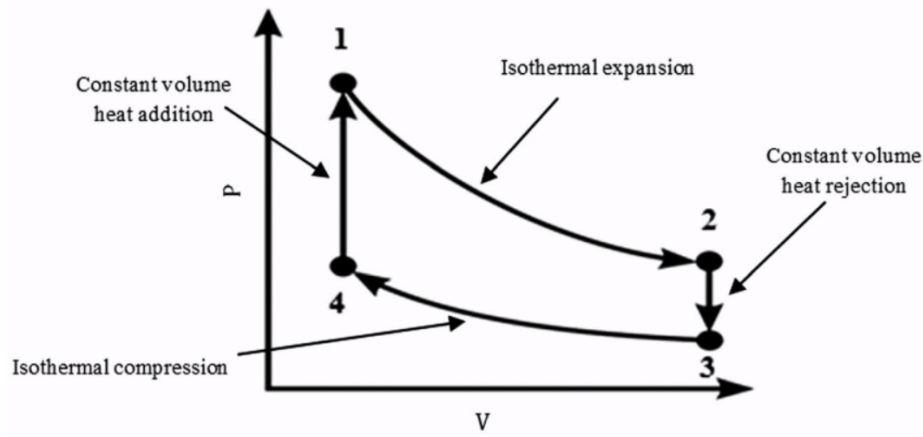


Figure 2 Ideal Stirling Cycle [2].

Stirling engines operate on a wide variety of working fluid, typically air, helium, hydrogen, nitrogen, or water vapors. The Stirling engine used in this system is a free-piston Stirling engine (FPSE). They are a well-defined sort of dynamic Stirling engines, and are known as the advanced type of traditional Stirling converters. They are maintenance-free and self-starting with a long operating life due to the lack of driving mechanisms in their structure. However, some challenges of these engines come from their dynamic characteristics. FPSEs are sensitive to their design parameters and are more complicated than the kinematic types [2].

There are four distinct spaces within a conventional Stirling engine, including expansion and compression spaces along with regenerator and dead volumes; these are shown in Fig. 3 [2].

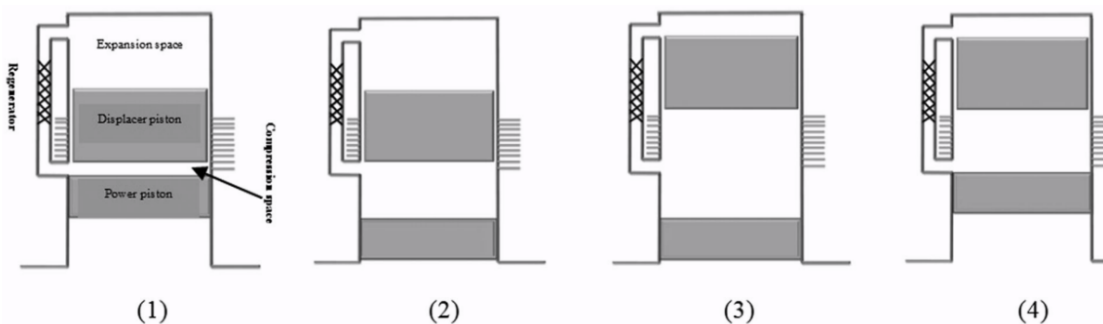


Figure 3 Working principle of the Stirling engine [2].

The heat flow model of the non-ideal Stirling cycle engine is shown in Fig. 4 [3].

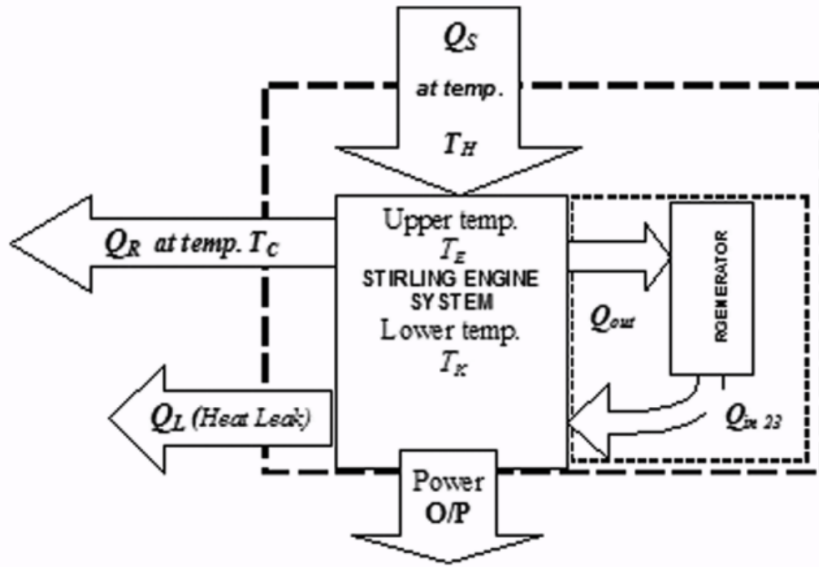


Figure 4 Heat Flow Model of Stirling Cycle Engine [3].

The combination of the solar dish-Stirling engine with thermal energy storage or hybridization lends itself to more continuous working of the system. It also reduces the dependency on the availability of solar energy. The solar dish-Stirling Engine designed to power the WashU machine shop will use battery storage to distribute energy after it gets dark. A factor of safety will also be applied, which is essential in Stirling solar dish systems to manage the high thermal flux and structural loads. As these systems achieve high efficiencies by concentrating solar energy, they require strict safety margins to prevent receiver overheating, mirror damage, and failure of the tracking or power conversion units during operation. An example of a solar dish-Stirling engine combined with battery storage schematic is shown in Fig. 5 [1].

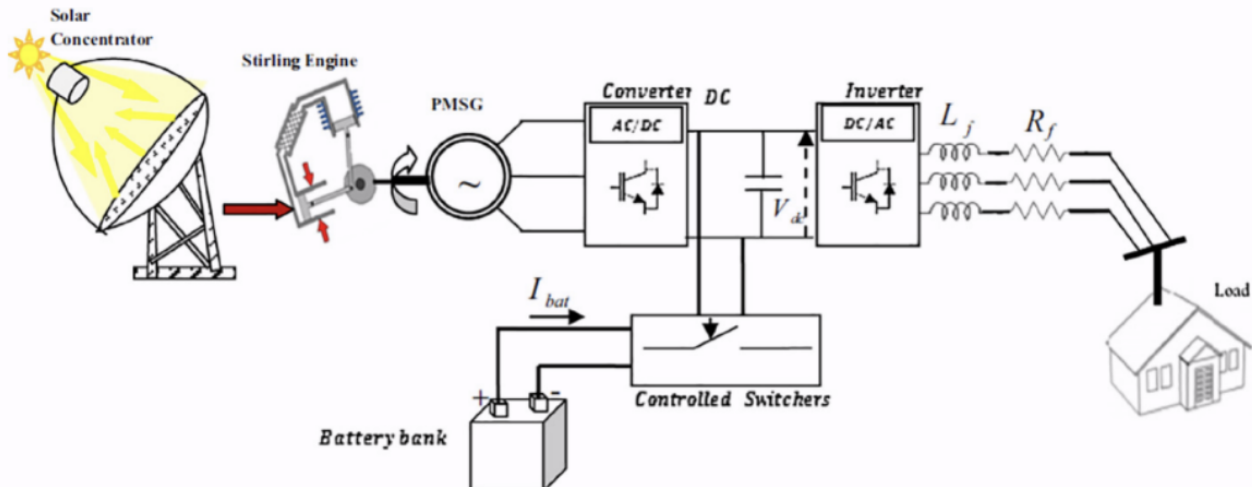


Figure 5 Solar dish-Stirling/PMSG system with battery storage [1].

Assumptions

- The working fluid is Helium, common in such devices for being non-toxic and non-flammable, which can be treated as an ideal gas (helium)[4].
- Kinetic and potential changes in the gas are neglected since the gas does not accelerate or move positions [1].
- The engine is ideal and operates in a steady state, with a constant mass of working fluid (no leakage), and mechanical losses and regenerator losses are neglected.
- The hot and cold volume processes are modeled as ideal isothermal processes, with heat addition determined by the solar input and receiver performance and heat rejection determined by convection[5].
- The optical and receiver system inefficiencies due to including reflectivity, tracking, and intercept losses are captured in η_{opt} and η_{rec} , respectively.
- The engine geometry is fixed, with a known maximum working volume V_{max} and a specified compression ratio r_v .
- Properties are evaluated at the T_{∞} for any given time of a day.
- Variable machine shop use is approximated as a maximum potential load, valid as it is the most conservative comparison for the model.

Boundary Conditions

- Average direct normal irradiance (DNI): $207.075 \frac{W}{m^2}$, and hourly data was compiled for the months of January and July[6].
- Dish diameter: $d_{\text{dish}} = 11m$ and receiver diameter: $d_{\text{rec}} = 0.44m$ for a concentration ratio of $C = A_{\text{dish}}/A_{\text{rec}} = 25$ [7, 8].
- Optical efficiency: $\eta_{\text{opt}} = 87\%$ [9].
- Receiver efficiency: $\eta_{\text{rec}} = 95\%$ [9].
- Maximum working volume: $V_{\text{max}} = 4L$ with a stroke to bore ratio of 0.84 for a $d = 16.2978cm$ and a $H = 19.1739cm$ [10, 11].
- Volume ratio: $r_v = 2$ and RPM = 400 [10, 12].
- Ambient temperature: $T_{\infty} = 30 - 42^{\circ}F$ in January and $T_{\infty} = 69 - 85^{\circ}F$ in July[13].
- Required net power output (machine shop demand): $\dot{W}_{\text{net}} = 50kW$.
- Gas constant for helium: $R = 2077 \text{ J/kg}\cdot\text{K}$.

- The cost of electricity in St. Louis is \$0.14 per kWh[14].

Equations

This section presents the governing equations in the order required to solve the system.

Ideal Gas

The working fluid obeys the ideal-gas law of Eq. 1 where P is pressure in $[Pa]$, v is specific volume in $[\frac{m^3}{kg}]$, R is the gas constant in $[\frac{J}{kgK}]$, and T is the temperature in $[K]$:

$$pv = RT \quad (1)$$

The ideal gas law is also valid in Eq. 2 since $v = \frac{V}{m}$ where V is volume in m^3 and m is mass in $[kg]$.

$$pV = mRT \quad (2)$$

The compression ratio is where both V_{min} and V_{max} are volumes in $[m^3]$.

$$r_v = \frac{V_{max}}{V_{min}} \quad (3)$$

Given V_{max} and r_v , the minimum volume is

$$V_{min} = \frac{V_{max}}{r_v} \quad (4)$$

For a fixed mass m in $[kg]$, the total pressure at state i in $[Pa]$ is

$$P_i = m \frac{RT_i}{V_i} \quad (5)$$

Solar Process

The solar power incident on the dish aperture is provided in Eq. 6 where $\dot{Q}$ is the heat energy entering the system in $[W]$, DNI is the direct normal irradiation in $[\frac{W}{m^2}]$, A_{dish} is the dish area in $[m^2]$ and η_{opt} is the optical efficiency of the dish.

$$\dot{Q}_{solar} = DNI A_{dish} \eta_{opt} \quad (6)$$

The useful thermal power delivered to the receiver is a fraction of the $\dot{Q}_{solar}$ where η_{rec} is the optical efficiency of the receiver.

$$\dot{Q}_{in} = \dot{Q}_{solar} \eta_{rec} = DNI A_{dish} \eta_{opt} \eta_{rec} \quad (7)$$

The geometric concentration ratio is

$$C = \frac{A_{dish}}{A_{rec}} \quad (8)$$

Isothermal work done by the boundary for a two-stroke engine is defined in Eq. 9 where $\frac{RPM}{60}$ is the revolutions per minute of the system converted into seconds.

$$\dot{W}_{iso} = m R_{gas} T \ln\left(\frac{V_2}{V_1}\right) \frac{RPM}{60} \quad (9)$$

An energy balance on the high-temperature expansion is as follows:

$$\dot{Q}_{in} - \dot{W}_{out} = 0 \quad (10)$$

By plugging in the known equations for work and energy at the high temp condition,

$$\dot{Q}_{in} = \eta_{opt} \eta_{rec} A_{dish} \text{DNI} = m R_{gas} T_H \ln\left(\frac{V_{max}}{V_{min}}\right) \frac{RPM}{60} \quad (11)$$

Provided a set solar input, mass, volume ratio, and RPM, T_H can be calculated in $[K]$.

$$T_H = \frac{\eta_{opt} \eta_{rec} A_{dish} \text{DNI}}{m R_{gas} \ln\left(\frac{V_{max}}{V_{min}}\right) \frac{RPM}{60}} \quad (12)$$

Cooling Process

At the low temperature isothermal process, the energy balance is as follows such that the work to compress the gas is equal to the heat rejected.

$$\dot{W}_{out} - \dot{Q}_{out} = 0 \quad (13)$$

Work is again defined as an isothermal process for the bottom half as seen in Eq. 9 such that $W_{iso} = m R_{gas} T_L \ln\left(\frac{V_4}{V_3}\right) \frac{RPM}{60}$.

The heat rejection is defined in Eq. 15 where $\dot{Q}_{out}$ is the heat rejected to the environment in $[K]$, h_{∞} is the heat transfer coefficient in $[\frac{W}{m^2 K}]$, and A_{cool} is the convective surface.

$$\dot{Q}_{out} = h_{\infty} A_{cool} (T_L - T_{\infty}) \quad (14)$$

Knowing the work and heat transfer definitions from Eqs. 9 and 15 respectively, the following expression can be determined.

$$m R_{gas} T_L \ln\left(\frac{V_4}{V_3}\right) \frac{RPM}{60} - h_{\infty} A_{cool} (T_L - T_{\infty}) = 0 \quad (15)$$

such that

$$T_L = \frac{T_{\infty}}{1 + \frac{R_{gas} T \ln\left(\frac{V_4}{V_3}\right) \frac{RPM}{60}}{h_{\infty} A_{cool}}} \quad (16)$$

The area, A_{cool} , is the cylinder's surface area in Eq. 17 where d is the cylinder's diameter and h is the cylinder's height, both in $[m]$.

$$A_{cool} = \pi d h \quad (17)$$

The cylinder dimensions are determined by the dimensionless bore ratio such that :

$$\text{BR} = \frac{d}{h} \quad (18)$$

By plugging in the bore ratio into the volume Eq. 19,

$$V = \frac{\pi d^2 h}{4} \quad (19)$$

One can derive the height in Eq. 20 and subsequently the diameter.

$$h = \left(\frac{4V}{\pi BR^2} \right)^{1/3} \quad (20)$$

The heat transfer coefficient can be calculated using forced convection Nusselt correlations. Flow is characterized by fluid properties and velocity in the Reynolds number in Eq. 21, evaluated at the T_{inf} temperature in [K] where ν is the kinematic viscosity in [$\frac{m^2}{s}$] and V is the external fluid velocity in [$\frac{m}{s}$].

$$Re = \frac{V_{air}d}{\nu} \quad (21)$$

The Nusselt correlation for flow over a cylinder is Eq. 22 where Pr is the fluid property Prandtl number, the ratio of momentum diffusivity to thermal diffusivity.

$$Nu = 0.3 + \left[0.62 Re^{1/2} Pr^{1/3} \left(1 + \left(\frac{0.4}{Pr} \right)^{2/3} \right)^{-1/4} \right] \left[1 + \left(\frac{Re}{282000} \right)^{5/8} \right]^{4/5} \quad (22)$$

From the Nusselt, the heat transfer coefficient can be calculated in Eq. 23

$$h = \frac{Nu k}{d} \quad (23)$$

Fan Power Consumption and Net Energy

The power consumption by the fan detracts from the net output of the system. Parasitic loads like fans in Stirling systems typically consume 3% of the total engine output [15]. The fan's power consumption is calculated below.

$$\dot{Q}_{fan} = 0.03 * W_{net} \quad (24)$$

The net energy to run the system is provided as

$$W_{net} = W_{out} - W_{in} - Q_{battery} = Q_{in} - Q_{out} - Q_{battery} \quad (25)$$

Machine shop power intake

The power supplied by the solar sterling engine is compared to the energy required to run the machine shop. The power per TRAK K3 knee mill is calculated using the equation below [16].

$$P_m \approx \sqrt{3} \cdot V \cdot I \cdot (pf \cdot \eta) \quad (26)$$

Where V is the voltage [V], I is the current [A], pf is the power factor, and η is the efficiency.

So a good design number per K3 mill is about 3 kW. For 8 of these TRAK K3 mills that live in the machine shop the final power is calculated below.

$$8 \times P_m \approx P_{mills} \quad (27)$$

The power of the lathes, which are a hybrid CNC/manual model we assume a range of 2-5 kW for a manual engine lathe and about 7-15 kW for small-medium CNC lathes. For the 5 lathes that are hybrid in the machine shop space.

$$5 \times P_l \approx P_{lathes} \quad (28)$$

The total shop peak load estimate:

$$P_{total} \approx P_{mills} + P_{lathes} \quad (29)$$

For the nature of the power usage, a demand factor of 0.8 will be applied (not all at full load) as well as a safety margin of 20-30%.

To calculate the excess heat produced by the system over the day, the trapezoidal rule can be applied.

$$E_{excess} = \sum_{t=9AM}^{4PM} (((W_{net}(t) \cdot 4) + \frac{1}{2}((W_{net}(t+1) - W_{net}(t)) \cdot N_{engines}) - 50kWhr), \quad (30)$$

Finally, the cost of energy usage in St. Louis is calculated by multiplying the load by the cost.

$$energy[kWhr] \times \frac{\$}{kWhr} = cost[\$] \quad (31)$$

Results and Discussion.

Figure 6 shows how the expected solar energy changes between summer and winter.

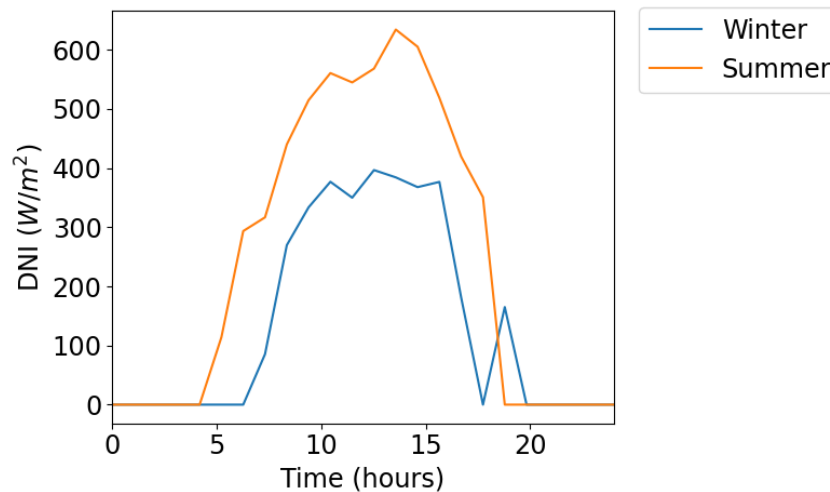
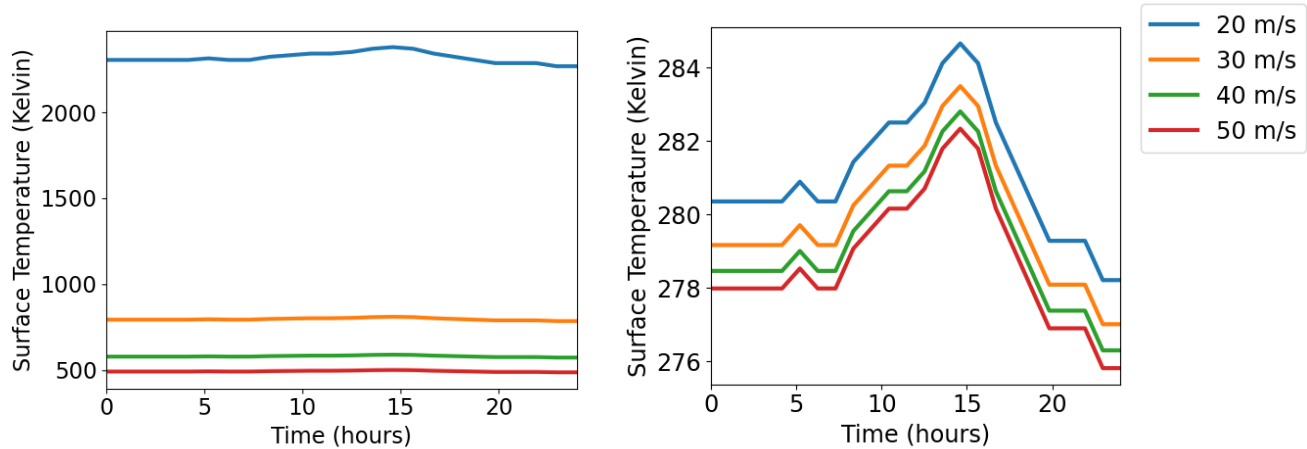


Figure 6 Solar Energy Input For Winter And Summer

The x-axis starts and ends at midnight, covering 24 hours. In both summer and winter cases, solar energy is highest during the day when there is more sunlight, and drops to zero during the night. The maximum solar energy is collected during the summer, while winter has the lowest expected energy input. This means that calculations should assume the conditions of winter so enough energy is collected even in sub-optimal scenarios.

Figure 7 shows how the surface temperature of the engine varies across the day for different fan speeds for both air and water.



(a) Surface Temperature with Air.

(b) Surface Temperature with Water

Figure 7 Low Surface Temperature Versus Time.

In Fig. 7a, considering air, as the fan speed increases, the expected surface temperature decreases. Having forced convection is necessary to keep the fluid temperature low between states 3 and 4. In this case, selecting a fan speed of 50 m/s is optimal since it reduces the surface temperature to 500 K. Further increases will decrease the temperature more, but will also be more expensive to operate. Notably, Fig. 7b considers the exact same operating conditions while using water as the crossflow fluid. It shows much lower surface temperatures than air for the same fan speed, suggesting that, when the heat transfer coefficient is improved, a less powerful fan can be used while still attaining reasonable surface temperatures.

Figure 8 shows how the net work changes for different hours of the day, and for different fan speeds.

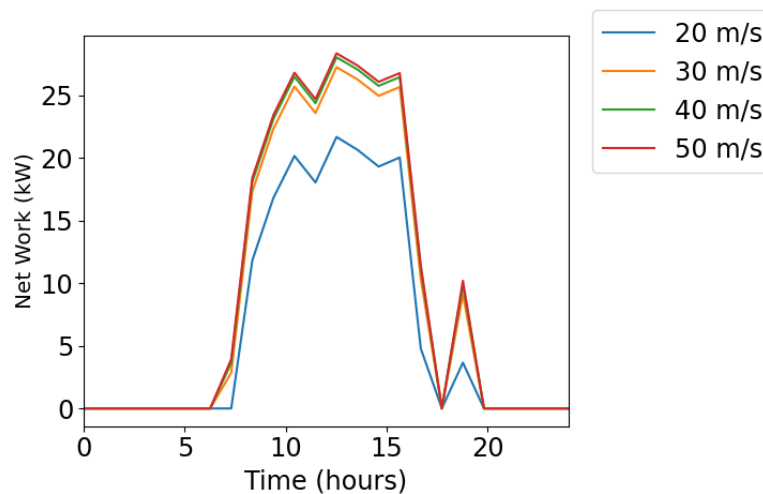


Figure 8 Net Work Versus Time of Day

The largest factor in the net work is the energy provided by the sun. This is multiplied by the area of one solar panel to determine the energy input. There is also the energy required during compression from states 3 to 4, and the energy required by the fan; though, these are very small compared to the solar energy. The net work increases with respect to fan speed, but the benefits become small at large speeds.

Like in Fig. 6, the expected net work drops to zero during the night time. This means that energy from battery supply will need to be used between, in the worst case, 4 PM and 9 PM. This can be supplied by excess energy generated by the engine between 9 AM and 4 PM. If it is assumed that there are 4 solar stirling engines, then the generated energy can be multiplied by 4. It is also necessary to subtract the 50 kW that will be required from the machine shop. Then, the excess energy can be determined by the trapezoid integration in Eq. 30 using $N_{engines} = 4$. The expected energy usage during night hours can be subtracted from the total excess energy to determine whether enough energy is produced to power the shop. With a fan speed of 50 m/s, the excess energy was determined to be 360 kWh. If it is assumed that the engine requires 204 kWh - based on 50 kW for the machine shop and fan energy from Eq. 33 for 4 hours - then more than enough energy is produced, with 156 kWh left over.

Table 1 shows how the performance varies between summer and winter at noon, assuming a fan speed of 50 m/s.

Table 1 Summer Versus Winter Engine Performance

	Jan. (Air, 0.5gHe)	July (Air, 0.5gHe)	Jan. (H2O, 0.5gHe)	Jan. (H2O, 5gHe)
T_H	8653 K	12397 K	8654 K	865 K
T_L	497 K	549 K	281 K	319 K
DNI	397 W/m ²	568 W/m ²	397 W/m ²	397 W/m ²
Q_{in}	31.1 kW	44.6 kW	31.1 kW	31.1 kW
W_{net}	28.4 kW	41.6 kW	29.1 kW	18.7 kW

The increased solar energy during the summer allows the engine to output more work. Using air as a coolant is compared with water to observe the effects of an increased heat transfer coefficient. Additionally, water is tested with 0.5 gHe and 5 gHe. Better heat transfer leads to a lower T_L , as observed by the change from 497 K in winter with air to 281 K in winter with 0.5 gHe. While this does not directly impact T_H , it does allow for a larger mass of Helium to be contained within the engine without dramatically affecting T_L . Notice how increasing the mass from 0.5 gHe to 5 gHe causes T_L for water to increase from 281 K to 319 K, but also causes T_H to decrease dramatically from 8654 K to 865 K. With the poor heat transfer of air, the improvement in T_H by increasing the mass of Helium would be greatly outweighed by a large increase in T_L . Therefore, while it is not technically feasible to immerse the engine in water, this shows that the performance of the engine is greatly improved by having an increased heat transfer coefficient. Some possible ways to implement better heat transfer may be to increase the surface area contact with air by adding fins, or add a coolant loop with a more suitable fluid than air. The cost saved per day by using the solar stirling engine is \$84 a day.

Sanity Check.

The thermodynamic results show the system is physically possible, because the net work output (W_{net})

scales proportionally with the solar irradiance changes observed between summer and winter. The net work output of $\approx 28.3\text{kW}$ is similar to many existing high output engines, with W_{net} values closer to 25kW [7]. While the calculated heater head temperatures (T_H) of 8,653 K to 12,397 K are mathematically consistent with the high concentration ratio and NSRDB [4] data, they exceed the melting point of standard receiver materials. This suggests that real-world applications would require higher mass flow or thermal spreading. Water has a higher heat transfer coefficient than air, but it also leads to corrosion and is not practical in many electrical applications[17]. One study found that adding fins to a surface increased the heat transfer coefficient by 19 %, so working fluids with better heat transfer capabilities and adding fins could improve the system's performance[18]. The system's cooling is realistic with an optimal fan speed of 50 m/s that maintains a surface temperature of 500 K, which aligns with established forced convection benchmarks for air-cooled industrial engines documented by NASA [10]. The calculated energy balance confirms that the system is accurately sized for the machine shop's peak demands, by providing a generated surplus of 368 kWh during daylight hours. After accounting for the 204 kWh required for evening operations, the leftover 156 kWh provides a critical safety margin to account for weather variability. This aligns with reliability standards for renewable energy storage [19]. The estimated total shop load of 50 kW is large in magnitude because the value is calculated when all the machines are actively being used. Realistically, the required load will be significantly lower on average because the number of machines being used varies. While the simplifications made in the model caused it to deviate some from literature values, it still serves as a helpful thermodynamic study and introduction into solar stirling engines.

Operations Manual.

The main function of the system is autonomous via a computer that controls dish tracking and engine operation. To optimize energy and maintain safety standards, a WashU machine shop specialist needs to flip the controlled switcher (shown in Fig. 5) around 4 pm . When the switcher is flipped, the machine shop (the load in Fig. 5) will draw electricity from the battery bank rather than directly from the Stirling engine. The battery bank is necessary because the solar dish system has varying electricity generation. When the solar dish produces more energy than the estimated peak load of 50 kW, the excess electricity gets automatically stored in the battery for later use.

Financial and Environmental Impact Analysis.

Solar energy is a clean energy source that reduces fossil fuel consumption and decreases environmental pollutants in the form of greenhouse gases. Among solar-based power generation systems, Solar dish Stirling systems have demonstrated particularly high efficiency, converting nearly 30% of incident solar radiation into electrical energy. These systems are expected to surpass parabolic trough technology by generating electricity at lower cost and higher efficiency, and they operate as self-contained power generators.

Dish Stirling systems offer numerous advantages. Technically, they achieve high thermal efficiency, deliver constant power output, and can accept any heat source as input with very little noise [1]. Financially, they benefit from high power densities and strong potential for long-term, low-maintenance operation [20]. Their modular nature further enables smaller, distributed applications, such as

supplying power to a specific area within a building. From an environmental standpoint, Solar dish Stirling engine systems require very little water for cooling, unlike conventional steam-based solar technologies. However, they do require significant raw materials for manufacturing.

However, deployment at a semi-large scale, such as a proposed installation near the WashU campus to capture direct normal irradiance (DNI), introduces certain tradeoffs. The addition of battery storage, while necessary to address supply variability, increases environmental burdens through the mining of raw materials and the use of toxic components, while also raising initial capital costs. This need for storage arises from the intermittent nature of solar energy, which produces fluctuating electricity output and creates mismatches between supply and demand. Despite saving \$84 a day, the cost of implementation for similar 25kW systems like ADI's Dual Shell Stirling Engine is \$25,000[21]. This means it would take 298 days to recoup the estimated implementation cost. While the initial cost is high, operation afterward would save money, recouping the cost in less than a year and saving \$30,000 of electricity costs in the next calendar year. During periods of surplus generation, energy storage systems are critical for capturing excess energy so that it can later be delivered in the required form, whether as electricity, thermal energy, or fuel, thereby ensuring reliability and continuity of the energy supply [19].

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Appendix.

1 Acknowledgment of AI Resources

AI tools were used in looking for literature resources.

Cited AI prompts

- Copilot, answer to "Can you help me find articles related to solar-Stirling engine environmental impacts?" Copilot, April 25, 2026.
- Copilot, answer to "Can you help me find resources related to lathes and mills of this size and load (provided calculated values)?" Copilot, April 25, 2026.
- Copilot, answer to "Can you help me revise my writing?" Copilot, April 28, 2026.
- Copilot, answer to "Can you help me convert these calculations into a LaTeX format?" Copilot, April 29, 2026.

2 Calculations.

Machine shop power intake

The power per TRAK K3 knee mill is calculated below. With the approximation of spindle motor at 3 hp, and the machine power at 220 V, 3 phase, and 8.5/4.25 A [16].

$$P \approx \sqrt{3} \cdot V \cdot I \cdot (\text{pf} \cdot \eta) \approx 1.732 \cdot 220 \cdot 8.5 \cdot 0.8 \approx 2.6 \text{ kW} \quad (32)$$

So a good design number per K3 mill is about 3 kW. For 8 of these TRAK K3 mills that live in the machine shop the final power is shown below.

$$8 \times 2.6 \text{ kW} \approx 21 \text{ kW}$$

The power of the lathes, which are a hybrid CNC/manual model we assume a range of 2-5 kW for a manual engine lathe and about 7-15 kW for small-medium CNC lathes. For the 5 lathes that are hybrid in the machine shop space.

$$5 \times 4\text{--}7 \text{ kW} \approx 20\text{--}35 \text{ kW}$$

The total shop peak load estimate:

$$P_{\text{total}} \approx 21 + 25 \approx 46 \text{ kW}$$

For the nature of the power usage, a demand factor of 0.8 will be applied (not all at full load) as well as a safety margin of 20-30%.

Fan Power Consumption

Parasitic loads like fans in Stirling systems typically consume 3% of the total engine output [15]. The power consumed by the fan is calculated below.

$$\dot{Q}_{fan} = 0.03 * W_{net} = 30 \text{ kW} * 0.03 = 900 \text{ W} \quad (33)$$

3 Code.

```
1
2 import scipy.optimize as opt
3 from matplotlib import pyplot
4 import numpy
5 from CoolProp.CoolProp import PropsSI
6 import csv
7
8 CR = numpy.linspace(2, 30, 50)
9
10 DNI = []
11 SolarDiam = 11
12 Ad = numpy.pi * (SolarDiam/2)**2
13 Ar = .12 * (0.01)**2
14 Cr = Ad/Ar
15 eff_opt = 0.87
16 eff_rec = 0.95
17 Vmax = 0.004
18 rv = 2
19 m = 1
20 Tinf_Far = numpy.array([34, 34, 34, 34, 34, 35, 34, 34, 36, 37, 38, 38, 39, ...
21                        41, 42, 41, 38, 36, 34, 32, 32, 32, 30, 30,
22                        71, 71, 70, 70, 69, 69, 70, 73, 76, 79, 81, 82, 84, ...
23                        85, 85, 85, 84, 84, 83, 80, 78, 77, 75, 72
24                        ])
25 Tinf = (Tinf_Far + 459.67) * 5/9
26 # print(Tinf)
27 hinf = 25
28 Ac = 1
29 Wnet = 1
30 Peff = 1
31 L = 1
32 k = 26.3 * 10**-3
33 v = 15.89 * 10**-6
34 alpha = 22.5 * 10**-6
35 Pr = 0.707
36 R = 2077
37
38 # 4 liters
39 # D = .183
40 H = 0.1917
41 D = 0.1629
42 Aengine = D * numpy.pi * H
43
44 # V = numpy.linspace(0, 10, 100)
45 V = 10
46
47 # get surface temperature from natural convection
48 Tsurf = []
49 for n, tinf in enumerate(Tinf):
50     v = PropsSI( V , T , tinf, P , 101325, Air )
51     k = PropsSI( conductivity , T , tinf, P , 101325, Air )
52     p = PropsSI( D , T , tinf, P , 101325, Air )
53     cp = PropsSI( C , T , tinf, P , 101325, Air )
54     alpha = k / (p*cp)
```

```

53     Pr = v / alpha
54     Re = V * D / v
55     Tsurf.append(tinf / (1 - ...
                    (R*numpy.log(rv)) / (0.027*Re**0.805*Pr**(1/3) * (k/D) * Aengine)))
56 Tsurf = numpy.array(Tsurf)
57
58 AllDNIData = []
59 for _ in range(24*2):
60     AllDNIData.append([])
61
62 line = 0
63 for row in csv.reader(open( DNI.Data.csv ), excel ):
64     line += 1
65     if line < 4:
66         continue
67
68     m = 0
69     month = int(row[1])
70     if month != 1 and month != 7:
71         continue
72     if month == 1:
73         m = 0
74     elif month == 7:
75         m = 1
76
77     hr = int(row[3])
78     dni = float(row[5])
79     AllDNIData[hr+m*24].append(dni)
80
81 for hr in AllDNIData:
82     DNI.append(numpy.mean(hr))
83
84 DNI_temp = []
85 for n, hr in enumerate(DNI):
86     i = n + 6
87     if i ≥ 48:
88         i -= 48
89     DNI_temp.append(DNI[i])
90 DNI = numpy.array(DNI_temp)
91
92 # solar input and reciever
93 Qsol = DNI*Ad*eff_opt
94 Qin = Qsol*eff_rec
95 C = Ad/Ar
96 qrec = DNI*C*eff_opt
97
98 eff = 50 * 1000 / Qin
99
100 Vmin = Vmax/rv
101 hours = numpy.linspace(0, 24, 24)
102
103 def PlotWorkAndTemp():
104     # pyplot.plot(hours, W[0:24])
105     pyplot.plot(hours, Qin[0:24])
106     # pyplot.plot(hours, Qout[0:24])
107     a1 = pyplot.twinx()
108     a1.plot(hours, Tsurf[0:24], g )

```

```

109     a1.plot(hours, Tinf[0:24], r )
110 def PlotDNI():
111     pyplot.plot(hours, DNI[0:24], label= Winter , fontsize= xx-large )
112     pyplot.plot(hours, DNI[24:48], label= Summer , fontsize= xx-large )
113     pyplot.title( DNI )
114     pyplot.legend(loc=(1.05, 0.65), fontsize= xx-large )
115     pyplot.tick_params( both , labelsize= xx-large )
116 def PlotQin():
117     pyplot.figure()
118     pyplot.plot(hours, DNI[0:24], label= Winter )
119     pyplot.plot(hours, DNI[24:48], label= Summer )
120     # pyplot.title( $Q_{in}$ )
121     pyplot.xlabel( Time (hours) , fontsize='xx-large')
122     pyplot.ylabel( DNI $(W/m^2)$ , fontsize= xx-large )
123
124     pyplot.xlim((0, 24))
125
126     pyplot.legend(loc=(1.05, 0.8), fontsize= xx-large )
127     pyplot.tick_params( both , labelsize= xx-large )
128     pyplot.tight_layout()
129 # 85$ / day
130 # def PlotNonIdealWork():
131 #     # wnets = w12 - w34s
132 #     # wnet = w12 - w34
133 #     # pyplot.plot(hours, wnets[0:24])
134 #     # pyplot.plot(hours, wnet[0:24])
135 #     # pyplot.plot(hours, m[0:24])
136 #     # pyplot.ylim((0, 60000))
137
138 def PlotQsolar():
139     R_helium = 2077.1
140     m = 0.0005
141     # m = 0.005
142
143     H = 0.1917
144     D = 0.1629
145     Aengine = D * numpy.pi * H
146     f = (600 / 60) / 2
147
148     Q12 = Qin
149     T1 = Q12 / (R_helium * numpy.log(rv))
150
151     for vel in [20, 30, 40, 50]:
152         # get surface temperature from natural convection
153         Tsurf = []
154         for tinf in Tinf:
155             fluid = Air
156             v = PropsSI( V , T , tinf, P , 101325, fluid)
157             k = PropsSI( conductivity , T , tinf, P , 101325, fluid)
158             p = PropsSI( D , T , tinf, P , 101325, fluid)
159             cp = PropsSI( C , T , tinf, P , 101325, fluid)
160             alpha = k / (p*cp)
161             Pr = v / alpha
162             Re = vel * D / v
163             # print(Pr)
164             Nu = 0.3 + (0.62 * Re**(1/2) * Pr**(1/3) * (1 + ...
                    (0.4/Pr)**(2/3))**(-1/4)) * (1 + (Re/282000)**(5/8))**(4/5)

```

```

165     # print(Nu)
166     h = Nu * k / D
167     # print(h)
168     # Tsurf.append(tinf / (1 + ...
        f*(m*R_helium*numpy.log(1/rv))/(h*Aengine)))
169     # print(Tinf)
170     Tsurf.append(tinf*h*Aengine / (h*Aengine + ...
        m*R_helium*numpy.log(1/rv)*400/60))
171 Tsurf = numpy.array(Tsurf)
172
173 P3 = m * R_helium * Tsurf / Vmax
174 Q34 = m * R_helium * Tsurf * numpy.log(1/rv) * f
175
176 # v = PropsSI ( V , T , (tinf+Tsurf)/2, P , 101325, Air )
177 # k = PropsSI ( conductivity , T , (tinf+Tsurf)/2, P , 101325, Air )
178 # p = PropsSI ( D , T , (tinf+Tsurf)/2, P , 101325, Air )
179 # cp = PropsSI ( C , T , (tinf+Tsurf)/2, P , 101325, Air )
180 # alpha = k / (p*cp)
181 # Pr = v / alpha
182 # Re = vel * D / v
183 # Nu = 0.3 + (0.62 * Re**(1/2) * Pr**(1/3) * (1 + ...
        (0.4/Pr)**(2/3))**(-1/4)) * (1 + (Re/282000)**(5/8))**(4/5)
184 # h = Nu * k / D
185
186 # Q34 = h * Aengine * (Tsurf-Tinf) * f
187 W12 = Q12
188 W34 = -Q34
189 Wnet = W12 - W34 - 1000
190 for n, v in enumerate(Wnet):
191     if v < 0:
192         Wnet[n] = 0
193
194 # Wnet = max(Wnet, 0)
195 # numpy.max
196
197 pyplot.plot(hours, Wnet[0:24]/1000, label=str(vel) + ' m/s ')
198 # pyplot.plot(hours, Tsurf[0:24], label=str(vel) + ' m/s ', linewidth=3)
199
200 totalW = 0
201 for n in range(7):
202     totalW += (Wnet[n+9] * 4) + 0.5 * ((Wnet[n+10]-Wnet[n+9]) * 4) - 50000
203 # print(totalW)
204 # print(total[12], totalW[12+24])
205 totalW -= (50000+1000)*4
206 print(totalW)
207
208 # print(Q12[12]/(f*m*R_helium*numpy.log(rv)), ...
        Q12[12+24]/(f*m*R_helium*numpy.log(rv)))
209 # print(DNI[12], DNI[12+24])
210 # print(Qin[12])
211 # print(Wnet[12], Wnet[12+24])
212 # print(totalW)
213
214 # print(Tsurf[12], Tsurf[12+24])
215
216
217 # # get surface temperature from natural convection

```

```

218 # Tsurf_nat = []
219 # for n, tinf in enumerate(Tinf):
220
221 #     v = PropsSI( V , T , tinf, P , 101325, Air )
222 #     k = PropsSI( conductivity , T , tinf, P , 101325, Air )
223 #     p = PropsSI( D , T , tinf, P , 101325, Air )
224 #     cp = PropsSI( C , T , tinf, P , 101325, Air )
225 #     alpha = k / (p*cp)
226 #     Pr = v / alpha
227
228 #     def nat_convection(x):
229 #         return tinf - x + (f*m*R_helium*x*numpy.log(rv)/Aengine) * (D/k) ...
#         * (0.6 + ( 0.387*( (2*9.81*(x-tinf)*D**3)/(v*alpha*(x+tinf)) )**(1/6) ...
#         )/( (1 + (0.559/Pr)**(9/16))**(8/27) ) )**(-2)
230
231 #     Tsurf_nat.append(opt.root(nat_convection, 500).x)
232
233 #     # Tsurf.append
234
235 #     # Nu = 0.3 + (0.62 * Re**(1/2) * Pr**(1/3) * (1 + ...
#     (0.4/Pr)**(2/3))**(-1/4)) * (1 + (Re/282000)**(5/8))**(4/5)
236 #     # h = Nu * k / D
237 #     # Tsurf.append(tinf / (1 - f*(m*R_helium*numpy.log(rv))/(h*Aengine)))
238 # Tsurf = numpy.array(Tsurf)
239
240 pyplot.xlabel( Time (hours) , fontsize= xx-large )
241 # pyplot.ylabel( Surface Temperature (Kelvin) , fontsize= xx-large )
242 pyplot.ylabel( Net Work (kW) , fontsize= x-large )
243
244 # pyplot.plot(hours, Tsurf_nat[0:24], label= Natural Convection )
245 pyplot.xlim((0,24))
246 pyplot.tick_params( both , labels= xx-large )
247
248 pyplot.legend(loc=(1.05, 0.65), fontsize= xx-large )
249 pyplot.tight_layout()
250 # pyplot.plot(P3, W12-W34)
251 # pyplot.plot(P3, W12)
252 # pyplot.plot(P3, W34)
253
254
255
256 # pyplot.plot(hours, Qs[0:24])
257
258
259 # print(Qs)
260 # pyplot.plot([0, 24], [(W12)/1000, (W12)/1000], label= $Q_{in}$ )
261 # pyplot.plot([0, 24], [(W12-W34)/1000, (W12-W34)/1000], label= $W_{net}$ )
262 # pyplot.plot(hours, Qin[0:24]/1000, label= $Q_{solar}$ )
263 # pyplot.xlim((0, 24))
264
265 # pyplot.ylabel( Energy Rate (kW) , fontsize= x-large )
266 # pyplot.xlabel( Time (hours) , fontsize= x-large )
267 # pyplot.legend(fontsize= x-large )
268 # pyplot.tight_layout()
269
270
271 # pyplot.plot(m, (W12-W34)/m)

```

```
272     # pyplot.plot(hours, Qout[0:24])
273
274     # PlotWorkAndTemp()
275     PlotQin()
276     # PlotNonIdealWork()
277     # PlotQsolar()
278     pyplot.show()
```